Cells and Gauge

File: gauge.kicad_sch

3V3 Rail

File: power.kicad_sch

MCU

File: mcu.kicad_sch

USB Port 1

IRQD
IRQ_N
EN_5V
EN_5V1
ADCIN1
P1_LDO_3V3
ADCIN2
GND
LDO_3V3D
P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQD
IRQ_N
EN_5V
EN_5V2
ADCIN1
GND
ADCIN2
LDO_3V3D
GND

File: usb_port.kicad_sch

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: ADCIN1 tied to its LDO_3V3, ADCIN2 to GND -> I2C 0x20
Port 2: ADCIN1 and ADCIN2 to GND -> I2C 0x21
Same SDA/SCL bus (CC2340 I2C0, 100 kHz): gauge BQ40Z50-R2 0x0B

MONOLITH PRODUCTION BOARD - SHEET INDEX
2. Cells and Gauge: welded 4S pack wires (28 AWG sense, 1N/4P Kelvin, 18 AWG PACK), BQ40Z50-R2 gauge, protection FETs, BQ294700 + F102 second-level protection, gauge SMBus on the GAUGE Tag-Connect (MCU sheet)
3. 3V3 Rail: TPS62177 always-on 3V3
4. MCU: CC2340R5MODA, Tag-Connect TC2030-NL pads: SWD (J301) and GAUGE (J302)
5. USB Port 1 / 6. USB Port 2 (one sheet file): TPS25751D + TPD6S300 + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (cell 1 negative at the PACK- power wire; gauge VSS is the CELL_1N label)
Spec and open items: Production/README.md. Derived from the dev board rev A8 (DevBoard/, history in DevBoard/README.md).

REVISION LOG
P0 2026-10-17 Dev board rev A8 with the dev-only parts removed (Production/tools/make_production.py, tidy_production.py):
                sleep-current section, bench supply, cell holders, headers, buttons, status LED, test points, DNP options
                and OR links. Added pack wire pads, TC2030-NL SWD + GAUGE pads, training and probe pads (P0 is the first build).
                Author: Claude Code. Reviewer: none yet.
P0 2026-10-21 Cup enclosure: the pack pads become 7 plated slots for nickel tails (PACK+/- power, 1N/4P Kelvin
                on new nets CELL_1N / CELL_4P, CELL1-3); cover standoffs H101-H104 become carrier peg holes.
Dev board history (A0-A8): DevBoard/README.md and the DevBoard root sheet.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Sheet index, PD address straps
Monolith 100 W battery bank — production

Sheet: /
File: Monolith.kicad_sch

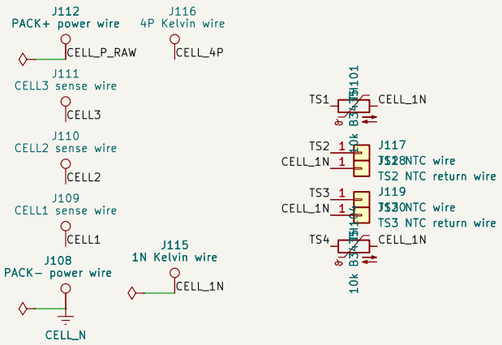
Title: Monolith production board — top level

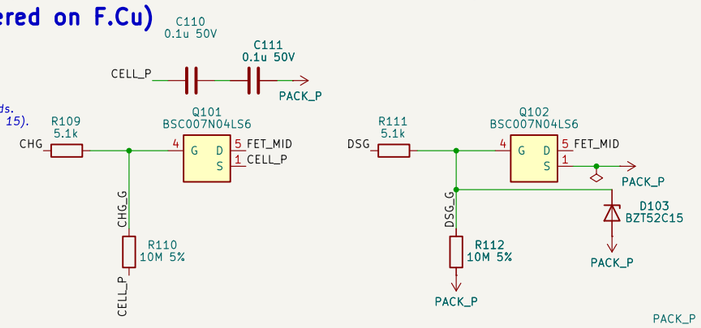
Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 1/6

PROTECTION FETs (back-to-back N-FETs, common drain)

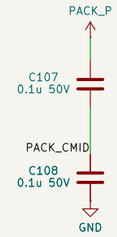
PACK SLOTS + WIRES (welded 4S BAK 65E, tails and wires soldered on F.Cu)

J108 PACK− / J112 PACK+: 18 AWG power wires in 1.5 mm holes. J115 1N / J116 4P: Kelvin wires from the same pads.
J109–J111: CELL1–3 28 AWG sense wires. USB unplugged, in order: PACK−, 1N, CELL1, CELL2, CELL3, 4P, PACK+ (note 15).

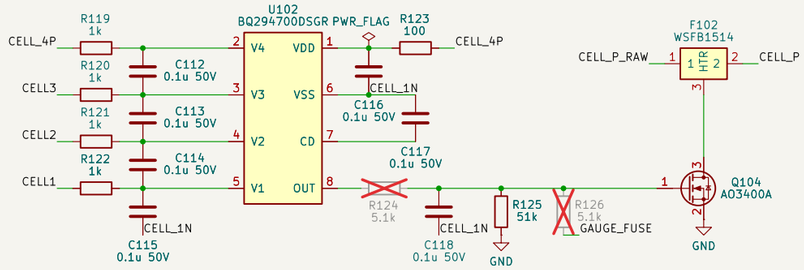




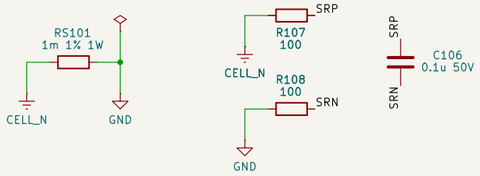
GAS GAUGE BQ40Z50-R2 (TI fig. 9-1)



SECOND-LEVEL PROTECTION BQ294700 + SCP (TI fig. 9-11)

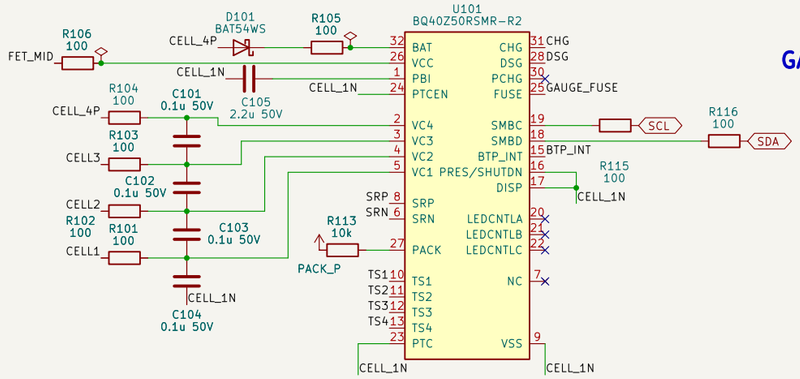


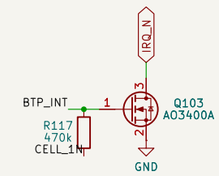
LOW-SIDE CURRENT SENSE



TEST + TRAINING PADS

Training: J113/J114 = learning-cycle load wires (20 AWG).
WAKE: bridge TP113 to TP106 (PACK_P).





NOTES
1. CELL_N = cell 1 negative at the PACK− power wire J108 (RS101). CELL_1N = gauge VSS, star-connected only at the 1N Kelvin wire hole J115 (TI SLUA660 2.3). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads: R107/R108/C106 at the IC. RS101 is a 1206 1 W 75 ppm alloy part (P0).
3. VC taps are separate thin traces from the sense-wire holes J109–J111. VC4 and the U102 V4/VDD filters are on CELL_4P, the 4P Kelvin wire hole J116 (before F102 and off the power wire): fuse and wire drop are not read as cell voltage.
4. TS1 / TS4 (front cells C1 / C4): SMD 0402 NTCs TH101 / TH104 on B.Cu in the gap-filler zones (smd-ntc.md). TS2 / TS3 (back cells): wired 10k NTCs on J117 / J118 (TS2) and J119 / J120 (TS3), taped to the cell at mid-height; insulated sensors only (cans 0–12.6 V above VSS, TS 2.1 V max). PTC, PTCEN, PRES, DISP tied to VSS per datasheet.
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_4P (cell top, before F102) via D101 + R105 (TI fig. 9-12).
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT onto the shared open-drain IRQ_N (CC2340 DIO11): IRQ_N is low while BTP_INT is high. Keep BTP disabled, or set BTP_POL = 1 with BTP_EN = 1, or IRQ_N is held low (review 4 D-2).
9. D103 clamps Q102 VGS at 15 V in a hard short. The dev board's C109/D102 inrush options are not on P0.
10. No wake button: the gauge leaves SHUTDOWN when PACK_P comes up (USB plugged in: TPS25751 dead-battery sink + BQ25756 charging with 3V3 down, USB sheet note 11). P0 fallback: bridge TP113 (WAKE, R118 4.7k from CELL_P) to TP106.
11. Gauge access is the GAUGE Tag-Connect J302 on the MCU sheet (SMBus + CHG_INH + NRST): BQStudio / EV2400 for training, golden image and calibration. J113/J114 = 20 AWG load wires (PACK_P/GND) for the learning-cycle discharge.
12. TP106 PACK_P and TP109 GND are 1.0 mm probe pads.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2), C107+C108 across PACK+/PACK− (sec. 9.2.2.1.5).
14. Second level (TI fig. 9-11): U102 BQ294700 (4.35 V/cell, CD 0.1u = 1–2 s) or the gauge FUSE pin turns Q104 on; F102's heater then blows its fuse, permanently opening the cell top. F102 is also the pack's only overcurrent fuse. Q104's source and gate pull-down R125 return to GND (cell negative through RS101): the heater current needs no long CELL_N trace.
15. Solder the pack with USB unplugged: PACK−, then 1N and the taps bottom-up, 4P, PACK+ last: a floating tap can read > 4.35 V, U102 trips after 1–2 s and F102 blows. Gauge data flash: COV Delay 1 s, Min Blow Fuse Voltage 10500 mV.
16. R124 / R126 are DNP from JLC (2026-10-24): with them off, neither U102 nor the gauge FUSE pin can turn Q104 on while the pack is wired up. Fit them by hand only after all seven pack voltages read right at the pads.

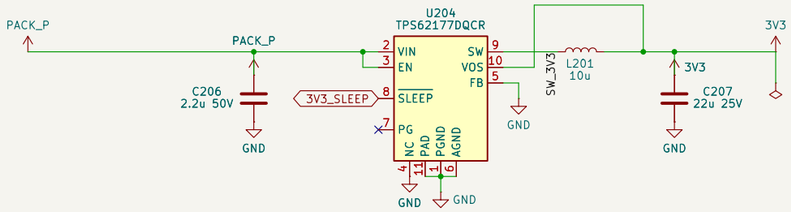
P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Welded 4S pack wires, BQ40Z50-R2 gauge/protector, protection FETs
Monolith 100 W battery bank — production

Sheet: /Cells and Gauge/
File: gauge.kicad_sch

Title: Monolith production board — cells and gauge

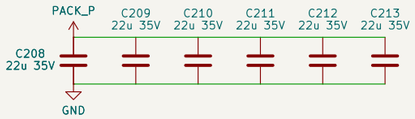
Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 2/6

3.3 V ALWAYS-ON RAIL (TPS62177, 1 MHz, sleep mode under MCU control)

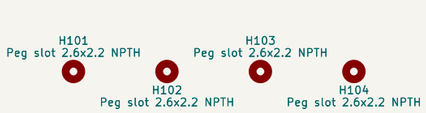


NOTES
1. PACK_P -> TPS62177 -> 3V3 (MCU + both TPS25751). Gauge BAT is NOT on this rail (TI: gauge must stay powered with the FETs off).
2. SLEEP low = sleep mode (internal 1.4 uA pull-down); CC2340 DIO21 (3V3_SLEEP) drives it, and R313 220k holds it high while U301 is in reset or unprogrammed.
3. PG is not used.
4. U204 pin 4 (NC) goes to GND, as TI recommends.
5. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
6. The dev board's sleep-current section (R201 R_MEAS, 2 x INA190, HT7550-1) is not on P0: PACK_LP is PACK_P. Measure sleep current on P0 with a meter in series with the PACK+ power wire (J112) before it is soldered; the gauge and U102 draw through the 4P Kelvin wire (J116) and are not included.
7. C208-C213 (6 x 22u 35V 1210) = the shared battery-side bulk for both chargers and both 5 V buck inputs (was one hybrid can per port). Place them between the two power stages on PACK_P, beside both RBAT pads (USB sheet note 5). Nothing on the board is taller than the 3.1 mm inductors.

PACK_P SHARED BULK (both chargers + both 5 V buck inputs): 6x 22u 35V 1210, between the two power stages beside the RBAT pads



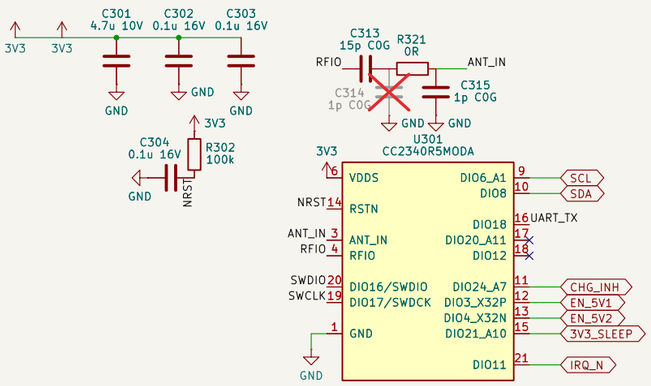
CARRIER PEG SLOTS (4x 2.6 x 2.2 mm NPTH for the cup carrier's 2 mm pegs; positions from mechanical/monolith_shell.py CUP_PEGS, placed by make_board.py --mech)



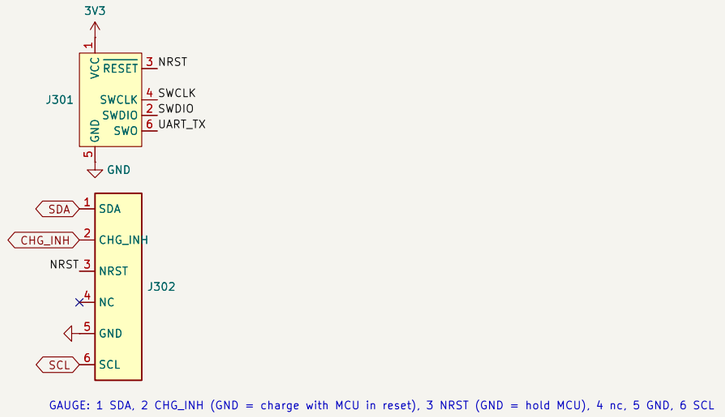
P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS62177 3V3 always-on rail
Monolith 100 W battery bank — production

Sheet: /3V3 Rail/		
File: power.kicad_sch		
Title: Monolith production board — 3V3 rail		
Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 3/6

BLUETOOTH MCU CC2340R5MODA (per TI SWRS349A fig. 9-1)

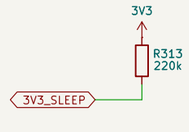


SWD (Tag-Connect TC2030-NL)

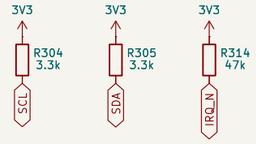


GAUGE: 1 SDA, 2 CHG_INH (GND = charge with MCU in reset), 3 NRST (GND = hold MCU), 4 nc, 5 GND, 6 SCL

3V3_SLEEP PULL-UP



I2C0 + IRQ PULL-UPS



3V3 PROBE PAD



NOTES
1. U301 antenna end at a board edge, ground 10–20 mm each side; keep-out in the footprint (fig. 9-2).
2. C301–C303 at the VDDS pads. R302 + C304 = power-on reset (SWRS349A fig. 9-1 R1/C92); no reset button.
3. C313 = CA1. R321/C314/C315 = Z61/Z60/Z62 pi match (TI: 0R, DNP, 1 pF); retune on the board.
4. No 32 kHz crystal: LFOSC calibrated against the 48 MHz crystal. DIO3/DIO4 are GPIOs.
5. Pins: DIO6 SCL, DIO8 SDA, DIO24 CHG_INH, DIO3/4 EN_5V1/2, DIO21 3V3_SLEEP, DIO11 IRQ_N.
   DIO18 = UART_TX (UART0TXD, mux 3) to J301 pin 6 for bring-up logs (2026-11-07). Spare (no-connect): DIO12, DIO20.
6. IRQ_N = TPS25751 IRQ x2 + gauge BTP (Q103), all open-drain. One pull-up: R314 47k.
   Firmware polls each source to find which.
7. Diamonds are PWR_FLAGs (ERC only, no part) where a rail is sourced through a passive part.
8. J301 = Tag-Connect TC2030-NL pads (no part fitted): XDS110 through a TC2030-IDC-NL cable. J302 (GAUGE) uses the same cable.
   Pin 6 (SWO on the Cortex pinout) carries UART_TX: tap it with a 3.3 V USB-UART RX. Don't plug the J302 EV2400 cable into J301 (pin 6 = SCL there). The debugger drives NRST. Bootloader: ROM bootloader on a blank part or the CCFG backdoor pin.
9. R313 220k holds 3V3_SLEEP high in reset (pins float; TPS62177 pulls down 1.4 uA typ).
   Costs 15 uA while driven low in S2. DIO21 is also the default ROM-bootloader pin:
   disable or move that trigger in the CCFG before the first flash (review 4 A-1).
10. R304/R305 SMBus pull-ups go to 3V3. Gauge work on J302 only with 3V3 up and U301 off the bus: hold NRST low (J302 pin 3)
    and ground CHG_INH (J302 pin 2) to keep charging enabled; an unpowered CC2340 can't take bus voltage (VDDS + 0.3 V).
11. DIO24 = CHG_INH: LOW allows charging. High or hi-Z stops both chargers
    (R417/R517 pull CE up), so reset/hang/unprogrammed stops charging.
12. Watchdog: CC2340 WDT runs in active, idle and standby and can't be stopped once enabled.
    GPIOs latch in standby; IRQ_N wakes it on an edge (SWRS349A 8.6, 8.11).
13. Temps: gauge DAStatus2 (cells, FETs); charger NTCs via TPS25751 'I2Cr' (BQ25756 REG0x37).
14. TP301 3V3 is a 1.0 mm probe pad.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
CC2340R5MODA, TC2030-NL SWD pads
Monolith 100 W battery bank — production

Sheet: /MCU/
File: mcu.kicad_sch

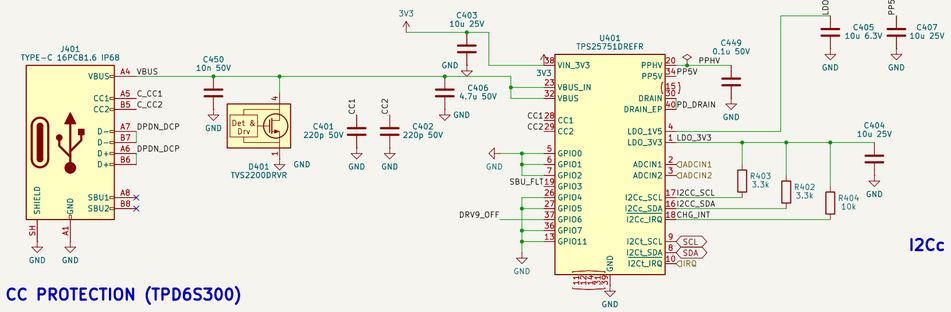
Title: Monolith production board — MCU

Size: A3	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 4/6

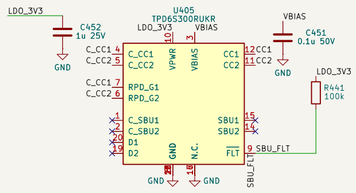
Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

USB-C CONNECTOR + PROTECTION

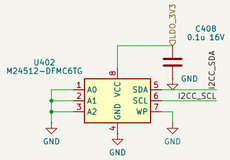
USB PD CONTROLLER TPS25751D



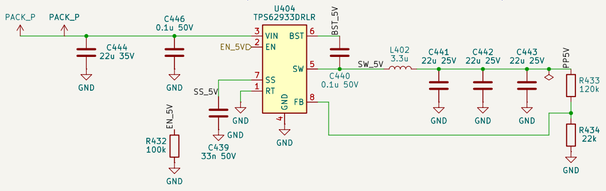
CC PROTECTION (TPD6S300)



I2Cc BUS: EEPROM + PULL-UPS



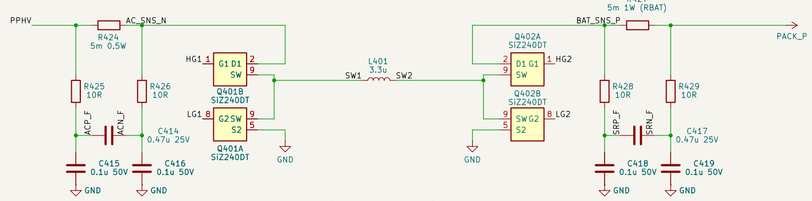
PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN



PROBE PADS (1.0 mm): VBUS PPHV PP5V REGN SW1 SW2 + GND

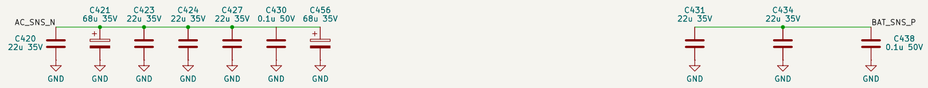


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)



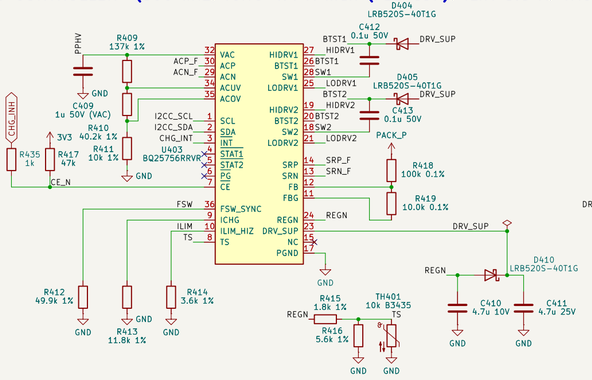
USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

BATTERY-SIDE LOCAL: 2 x 22u 35V 1210; shared PACK_P bulk C208-C213 (3V3 Rail sheet)



BQ25756 CONTROLLER (500 kHz, ICHG 4.4 A max (FEB cell), ILIM 5.6 A, VAC 4.1-22.5 V window)

9 V GATE DRIVE (DRV_SUP): fitted, off by default (loss cut 2)



GATE LINKS (OR, fitted) + SNUBBERS (DNP): P0 tuning hedge



NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
   port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 4, 5, 7, 11) tied to GND per datasheet (KiCad ERC flags this; expected).
   Never configure them as outputs. GPIO6 = DRV9_OFF: the 9 V gate-drive enable (note 17).
3. CC is protected by U405 (TPD6S300: CC short-to-VBUS OVP, 24 V DC, + IEC 61000-4-2 ESD; the TPS25751 CC pins are
   26 V abs max but only +-1 kV HBM). VPWR on LDO_3V3, VBIAS C451 0.1u 50V (>= 35 V needed). RPD_G1/G2 to
   C_CC1/C_CC2: dead-battery Rd while unpowered. Its SBU and D1/D2 channels are unused (no-connects).
   C401/C402 (220 pF) on the controller side: with the TPD6S300's 60-120 pF the total stays under 480 pF.
   D+ and D- are tied together at the connector (net DPDN_DCP: passive BC1.2 DCP); the TPS25751 does no BC1.2.
   No liquid detection and no SBU use on production (IP68 connectors, potted board): the connector SBU pins are
   no-connects; GPIO4, 5 and 7 are tied to GND (note 2). U405 FLT -> GPIO3 = Fault_Input_Event (SBU_FLT, R441):
   a CC short to VBUS forces error recovery.
   D401 = TVS2200 flat clamp on VBUS (TI's pick for the TPS25751, 28 V at 40 A 8/20 us); fitted as Maxinmicro TVS2200DRVR C55616459.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Low-profile bulk, all <= 3.1 mm (the inductor height). Two MLCCs of each bank plus C430/C438 at the FETs, the
   rest one row back. USB side: 4 x 22u 35V 1210 (C420 C423 C424 C427) + 2 x 68u 35V KEMET T593Y polymer tantalum
   (C421 C456, 2.0 mm): 133 uF worst case at 20 V, >= the BQ25756 80 uF minimum. Battery side: 2 x 22u 35V 1210 per port at
   the FETs; both ports, both TPS62933 inputs (C444, 22u 35V) and the six shared C208-C213 form one PACK_P bus:
   about 86 uF worst case, all MLCC (no leakage on the always-live bus). Production/tools/low_profile_sims.py C4/N1.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly: take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via D410, or from U406 (note 17).
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5nkBulkPd: the TPS25751D sink switch soft-start limits
   inrush (C x SS: about 170 uF effective x 0.47 V/ms max = 0.08 A at SS setting 0).
   Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH: R417 pulls CE up to 3V3, so charging needs the MCU (DIO24) to drive CHG_INH low through R435.
    Reset, WDT reset, the ROM bootloader or an unprogrammed MCU stop charging; CC2340 Standby keeps the pin state.
    A dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Gate drive: R444-R447 are 0R 0402 links at the gates (P0 hedge, no dev board): swap to 2.2-10R if SW1/SW2 ring;
    R460/C454, R451/C455 are DNP 0402 RC snubbers (3R3 + 33p). No turn-off diodes: BTST straight to its cap. Keep the
    gate traces short and wide (TI: >= 0.5 mm) with the FETs right at the BQ25756.
    Each leg is one Vishay SiZ240DT half-bridge (Q401/Q402: unit A = high side, unit B = low side, pin 9 = SW).
    40 V, 12.25/13.30 mOhm max at 4.5 V, Qg 6.9/6.5 nC. Worst case (20 V 5 A out from a 10 V pack) its fully-on
    high side dissipates ~1.8 W in one 3.3 x 3.3 package; needs the paddles' via arrays and the gap pad: the
    thermal throttle covers it. Buy from DigiKey/Mouser (JLC stock is near zero).
13. TP401 VBUS is a 1.0 mm probe pad. STAT1/STAT2/PG (U403 pins 4-6) are open-drain and unused (no pull-ups):
    status comes over I2C (TPS25751 I2Cr): set DIS_STAT_PINS = 1 in the BQ25756 configuration.
14. PP5V output: C441-C443 3 x 22u 1206 (about 33 uF effective) plus C407 10u at the PP5V pin. Each port's PP5V is
    its own TPS62933, a dedicated supply; USB PD cSrcBulk (10 uF min) applies, not the 120 uF cSrcBulkShared the
    TPS25751 table cites. Estimated 0.5 -> 3 A step undershoot about 190 mV (cap sims P3): measure it on P0.
15. L401 = Coilcraft XGL7030-332MEC 3.3 uH on the APS0730 pads (loss cut 3: consigned): >= 2.2 uH to 13 A, R412
    49.9k sets ~500 kHz. L/RSNS 0.66 uH/mOhm (TI: 0.05-1.26). P0 A/B: APS0730M2R2F on one port, XGL on the other.
16. RAC R424 is 0805 0.5 W (<= 0.16 W at 5.6 A), RBAT R427 1206 1 W (~0.5 W at 10 A). Both stay 5 mOhm, so ICHG/ILIM
    settings are unchanged. 0201 is used only where both nets stay <= 5.5 V.
17. 9 V gate drive (U406, loss cut 2): fitted, off by default; see the note beside it.



PO: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx

Monolith 100 W battery bank — production

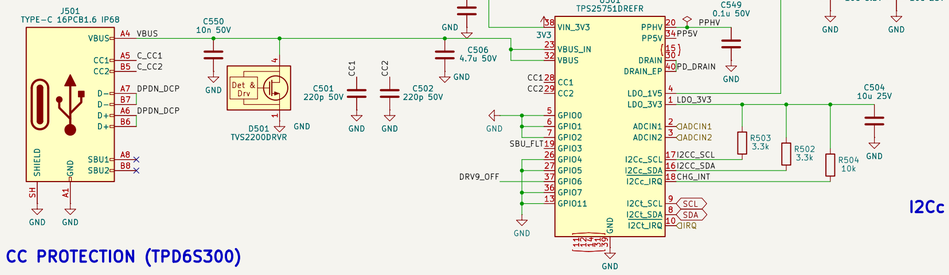
Sheet: /USB Port 1/
File: usb_port.kicad_sch

Title: Monolith production board — USB-C port

Size: A2	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 5/6

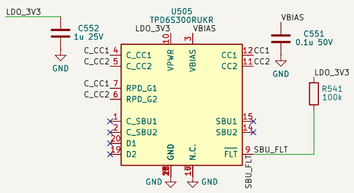
Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

USB-C CONNECTOR + PROTECTION

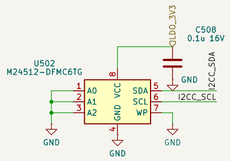


USB PD CONTROLLER TPS25751D

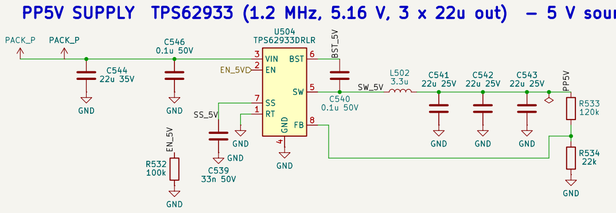
CC PROTECTION (TPD6S300)



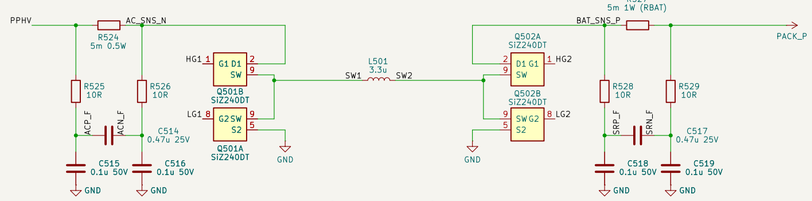
I2Cc BUS: EEPROM + PULL-UPS



PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN

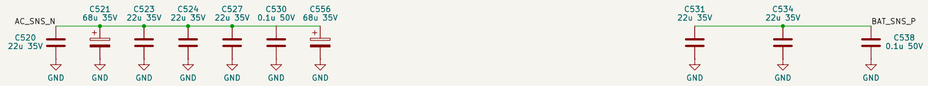


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)



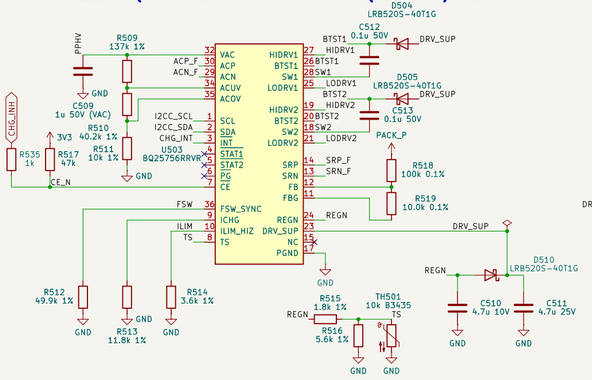
USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

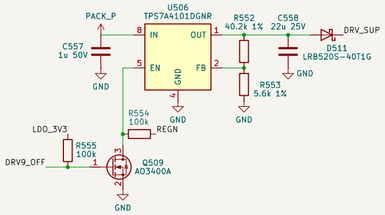
BATTERY-SIDE LOCAL: 2 x 22u 35V 1210; shared PACK_P bulk C208-C213 (3V3 Rail sheet)



BQ25756 CONTROLLER (500 kHz, ICHG 4.4 A max (FEB cell), ILIM 5.6 A, VAC 4.1-22.5 V window)

9 V GATE DRIVE (DRV_SUP): fitted, off by default (loss cut 2)





PROBE PADS (1.0 mm): VBUS PPHV PP5V REGN SW1 SW2 + GND



GATE LINKS (OR, fitted) + SNUBBERS (DNP): P0 tuning hedge



NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 4, 5, 7, 11) tied to GND per datasheet (KiCad ERC flags this; expected). Never configure them as outputs. GPIO6 = DRV9_OFF: the 9 V gate-drive enable (note 17).
3. CC is protected by U405 (TPD6S300: CC short-to-VBUS OVP, 24 V DC, + IEC 61000-4-2 ESD; the TPS25751 CC pins are 26 V abs max but only +-1 kV HBM). VPWR on LDO_3V3, VBIAS C451 0.1u 50V (>= 35 V needed). RPD_G1/G2 to C_CC1/C_CC2: dead-battery Rd while unpowered. Its SBU and D1/D2 channels are unused (no-connects). C401/C402 (220 pF) on the controller side; with the TPD6S300's 60-120 pF the total stays under 480 pF. D+ and D- are tied together at the connector (net DPDN_DCP: passive BC1.2 DCP); the TPS25751 does no BC1.2. No liquid detection and no SBU use on production (IP68 connectors, potted board): the connector SBU pins are no-connects; GPIO4, 5 and 7 are tied to GND (note 2). U405 FLT -> GPIO3 = Fault_Input_Event (SBU_FLT, R441): a CC short to VBUS forces error recovery. D401 = TVS2200 flat clamp on VBUS (TI's pick for the TPS25751, 28 V at 40 A 8/20 us); fitted as Maximicro TVS2200DRVR C55616459.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Low-profile bulk, all <= 3.1 mm (the inductor height). Two MLCCs of each bank plus C430/C438 at the FETs, the rest one row back. USB side: 4 x 22u 35V 1210 (C420 C423 C424 C427) + 2 x 68u 35V KEMET T593Y polymer tantalum (C421 C456, 2.0 mm): 133 uF worst case at 20 V, >= the BQ25756 80 uF minimum. Battery side: 2 x 22u 35V 1210 per port at the FETs; both ports, both TPS62933 inputs (C444, 22u 35V) and the six shared C208-C213 form one PACK_P bus: about 86 uF worst case, all MLCC (no leakage on the always-live bus). Production/tools/low_profile_sims.py C4/N1.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly: take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via D410, or from U406 (note 17).
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16. RAC R424 is 0805 0.5 W (<= 0.16 W at 5.6 A), RBAT R427 1206 1 W (~0.5 W at 10 A). Both stay 5 mOhm, so ICHG/ILIM settings are unchanged. 0201 is used only where both nets stay <= 5.5 V.
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Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — production

Sheet: /USB Port 2/		
File: usb_port.kicad_sch		
Title: Monolith production board — USB-C port		
Size: A2	Date: 2026-10-17	**Rev: P0**
KiCad E.D.A. 10.0.8		Id: 6/6